H_2O_2 Low Temperature Plasma Sterilizer
Installation Manual

PS-100X/200X

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Welcome use H_2O_2 Low Temperature Plasma Sterilizer. To maximize the performance of the equipment, please carefully read and comply with this manual for safety and operational purpose.

This manual is applicable to PS-100X/200X H_2O_2 Low Temperature Plasma Sterilizer.

Keep this manual perfectly during the service life of the equipment;
Make sure all updated information received has been included in this manual;
This manual must be transferred or handed over as part of the equipment whenever the equipment occupancy site is altered.

CONTENTS

1. CAUTIONS AND ATTENTIONS 2

1.1. ATTENTIONS 2
1.2. SAFETY SIGNS 4

2. EQUIPMENT INSTALLATION 4

2.1. EQUIPMENT INFORMATION 4
2.2. INSTALLATION REQUIREMENTS 4
2.3. EQUIPMENT IN-POSITION PROCESS 5

3. CHECKING AND RUNNING OF EQUIPMENT 6

3.1. DISMANTLE THE DOOR PANEL SUPPORT PLATE 6
3.2. CHECK THE VACUUM PUMP 7
3.3. CHECK WIRES AND PIPELINES 8
3.4. SUPPLY POWER TO THE EQUIPMENT AND CHECK ELECTRIC PARTS 8
3.5. CHECK CASSETTE PARTS 10
3.6. RUN THE PROCEDURE 11
3.7. CHECK THE BASKET 11
3.8. INSTALLING THE OUTER COVER 11

4. EQUIPMENT ACCEPTANCE 11

APPENDIX 1. INSTALLATION DIAGRAM OF PS-100/200 (SINGLE-DOOR) 13

APPENDIX 2. INSTALLATION DIAGRAM OF PS-100/200(DOUBLE-DOOR) 14

APPENDIX 3. LAYOUT DIAGRAM OF ELECTRIC BOX MAIN BOARD 15

APPENDIX 4. WIRING DIAGRAM OF ELECTRIC BOX 16

APPENDIX 5. TRANSFORMATION, RECTIFICATION AND POWER-ON SCHEMATIC 17

APPENDIX 6. ELECTRICAL CIRCUIT SKETCH 18

APPENDIX 7. WIRING DIAGRAM OF PUMP STARTUP BOX 19

5. APPENDIX 8. BOTTOM VIEW OF EQUIPMENT 21

6. APPENDIX 9. LEFT SIDE OF EQUIPMENT 22

1. Cautions and attentions

1.1. Attentions

Read carefully the following information while operating or maintaining the sterilizer. Please pay attention to these signs wherever they appear.

It indicates that please pay high attention.

It indicates that please obey or damage may happens on sterilizer.

It indicates that please strictly obey or human safety may be endangered.

Operators should always keep the following information in mind before operation.

Wherever the sign appears, please look up the relevant information in this manual to clarify the potential danger and take proper measures.

Anyone in any situation is not allowed to tear off any page of this manual. Please contact us promptly if this manual cannot help you with the sterilizer problems. Please make sure your manual is updated.

Please store this manual at ventilated and dry place preventing damps and heat.

The sterilizing agent contains H_2O_2 which could permeate into and temporarily skin. Please wear gloves that are made of latex or ethylene while handle the sterilizing agent. Handle the used cassettes properly. In any case of that your skin contacts the sterilizing agent; please wash it by plenty of water immediately for 5 minutes. If anyone accidently swallows the agent please drink plenty of water and turn to the doctor.

The following articles (including but not limited) mentioned are not applicable to this H2O2 low temperature sterilizer:

a. PS-100X:

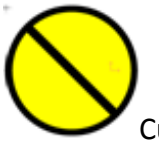
The stainless steel tubular instruments with inner diameter＜1mm and length＞500mm.

The Teflon tubular instruments with inner diameter＜1mm; length＞2000mm
PS-200X:
The stainless steel tubular instruments with inner diameter＜0.8mm and length＞300mm.
The Teflon tubular instruments with inner diameter＜1mm; length＞1000mm
b. Hygroscopic materials (i.e. wooden instrument, cellulose, cotton fabric, gauze, etc.);
c. Disposable instrument (i.e. disposable waterproof fabric, disposable surgical gown, etc.);
d. Liquid, cream, oil, and powder;
e. Half-dried items;
f. The tubular instruments with one sealed end;
g. Implants;
h. The instruments that could not bear vacuum pressure;
i. The instruments that are only allowed to be sterilized by pressure steam sterilizer or EO sterilizer;
j. The instruments that contain complicated construction e.g. sealed bearing;
k. The instruments that contain nylon surface;
l. The soft endoscopes with fossilized or damaged surface that their materials cannot be determined.
We do not make any explicit or implied guarantees or warranties for any damage caused by biogenetic or secondary injury and improper operation.
Please inquire our engineering department if you are not sure whether your instrument can be processed by this sterilizer.

All instruments must be thoroughly washed and dried, and properly packed by specified packing material or container before being put into chamber for sterilization.

To guarantee the performance of the sterilizer, it is necessary to carry out regular and preventive maintenance.

Any product adjustment or maintenance must be left to qualified staff. Improper maintenance or unauthorized installation of spare parts may cause personal injury.


Cut off all power supply of the sterilizer before maintenance.

1.2. Safety signs

On (power on) IEC 417 , No. 5007	Off (power off) IEC 417, No. 5008	Ground IEC 417 , No. 5017	Protective ground IEC 417, No. 5019
DC IEC417 , No. 5031	AC IEC417 , No.5032	Refer to the attached materials ISO3864 No.B.3.1	Prevent electric shock ISO3864, No. B.3.6
Prevent surface overheat IEC 417 , No. 5041	Refer to user manual YY0466 , No. 3.3	Keep dry YY0466 , No. 3.8	Temperature limitation YY0466 ,No. 3.11
Avoid direct sunlight YY0466,No. 3.6	Avoid high voltage IEC417 , No.5036	Fuse IEC 417, No. 5016	Keep in upright position GB/T191 ,No. 3

2. Equipment installation

2.1. Equipment information

Order number / Technical parameters	PS-100X	PS-200X
Sterilizing chamber size（LxWxH）mm	750x450x400	1250x450x400
Volume (L)	135	225
Overall dimension（LxWxH）mm	990x760x1600	1460x760x1600
Weight（Kg）	400	500
Power supply connection	380V 50Hz	380V 50Hz
Power（KW）	3	3.5

2.2. Installation requirements

1）Only indoor installation and usage are allowed

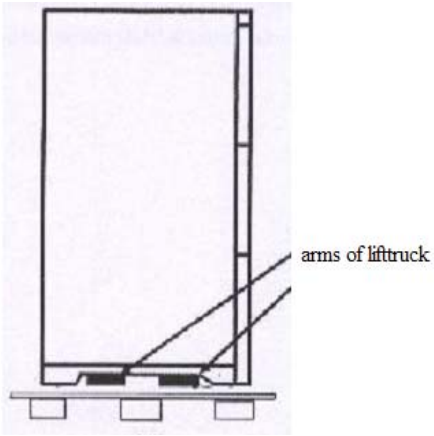
2）The input voltage is 380V, and the fluctuation tolerance range: 1 0 % (should be grounded).
3）Placement: Use caster moving equipment.
4）Working conditions
- Ambient temperature range: +10℃~+40℃。
- Relative humidity range: 30%~75%。
- Atmospheric pressure range: 70kPa ~106kPa。
5）Installation space ：
- Ground flatness 5mm/m。
- Smallest distance of two sides and back side of equipment away from the wall：100mm。
- PS-100X Distance between the equipment operating end and obstacles is larger than 1m。
- PS-200X Distance between the equipment operating end and obstacles is larger than 1.5m。
- Equipment maintenance space：larger than 1m on two sides.

2.3. Equipment in-position process

1）Check whether the wood box is complete before unpacking.
2）Unpack the wood box on flat open space.

The wood box must be always placed upwards; and do not damage the equipment.

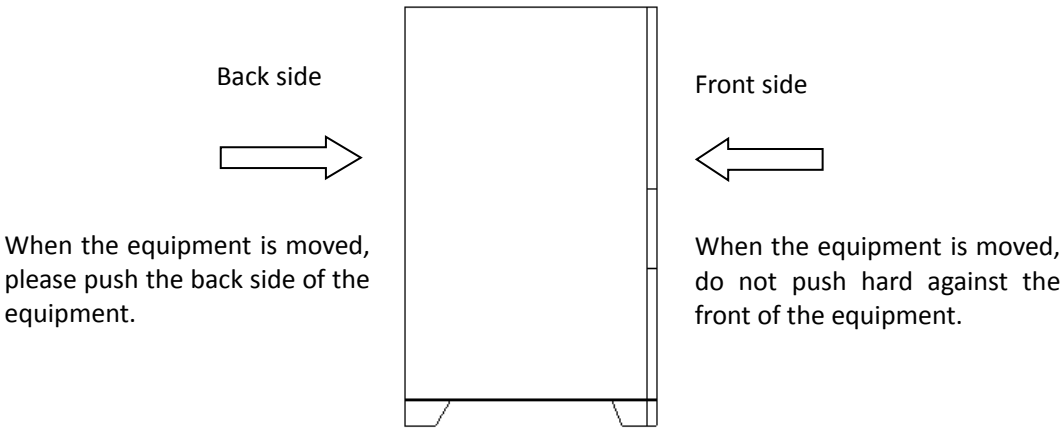
3）Remove the connecting piece between the equipment and the base of the wood box.
4）Use a pallet jack according to illustration, insert pallet jack arms from the bottom left side or bottom right side of the equipment box, until they come out about 200mm on the other side of the equipment. Lift the equipment by 200mm from the ground carefully. Attention should be paid to avoid electrostatic discharge when elevator arms and unloading equipment are inserted into the bottom of the equipment.



Position of elevator arms

5）Place the sterilizer on flat ground.

Moving equipment: if necessary, the equipment may be moved to the installation place from the unpacking place through handcarts by two persons.



Notes for moving the equipment

6）After the sterilizer is in position, rotate the feet on the bottom of the equipment clockwise to put them in complete contact with the ground. This will prevent the equipment from moving during operation. Please refer to Bottom Sketch of Equipment in Appendix VIII for the correct position of the feet.

7）After installation or reinstallation, let the equipment stand at least one hour before starting for the first time.

3. Checking and running of equipment

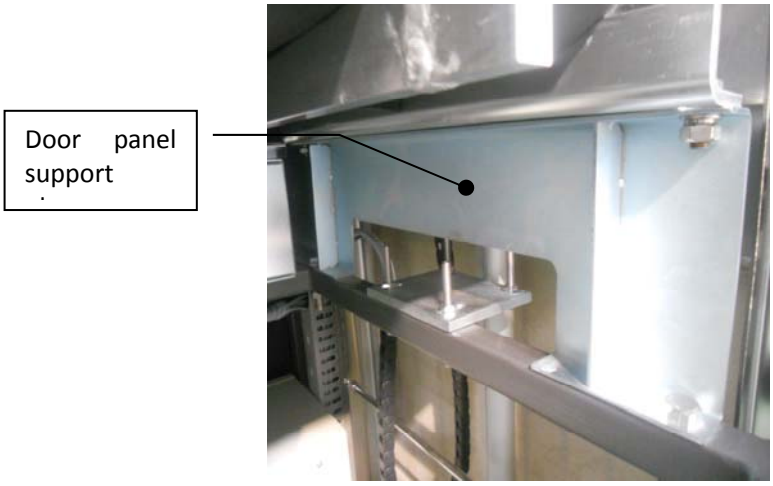
3.1. Dismantle the door panel support plate

To ensure that components of the product were not damaged during long-distance transport, we attached the door structure through a door panel support plate on the bottom of the automatic door. Therefore, the user has to dismantle the door panel support plate (save the door panel support plate and screws well after dismantling them) before normal usage, before startup will be allowed. The equipment can be debugged and used only after the door panel support plate is dismantled.

Do not start up the equipment before dismantling the door panel support plate!

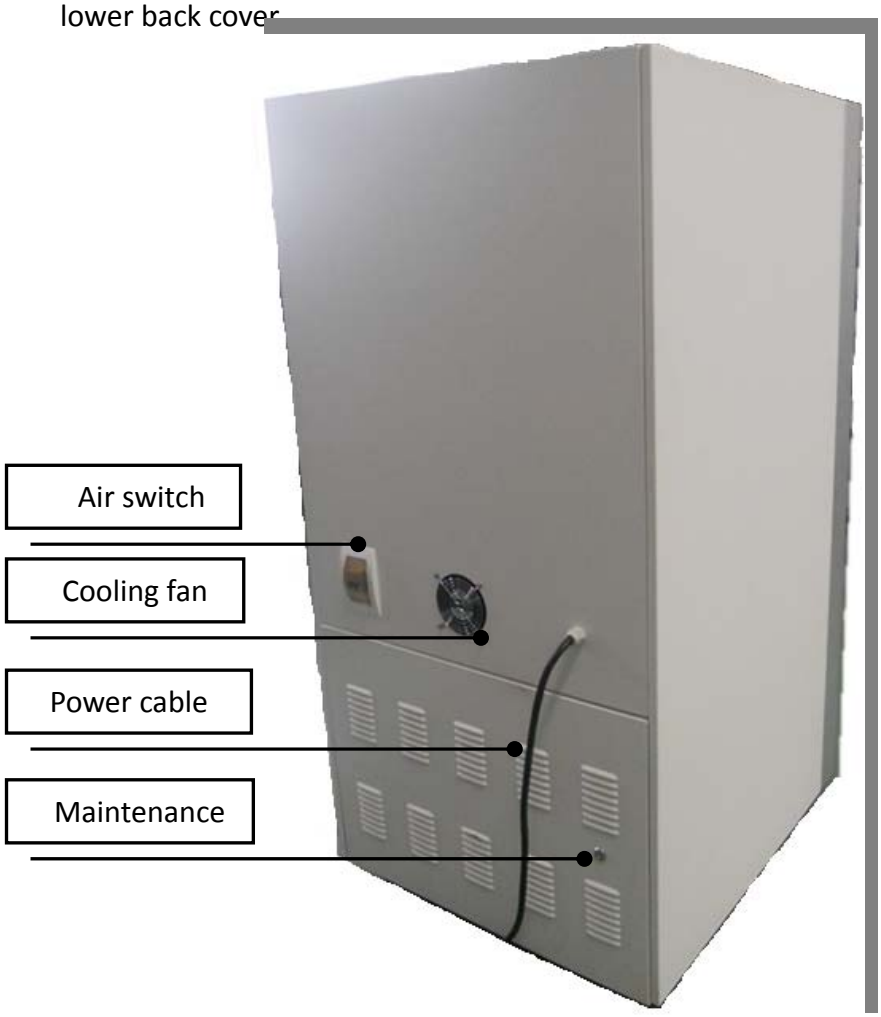
The specific dismantling method is as follows：

1) Remove the two screws which are located M5* 10mm on the lower bottom of the left cover (see Left View I of Equipment in Appendix IX), and take off the left cover. During dismantling, do not knock against the surface of the cover to avoid scratching the equipment.
2) Dismantle the door panel support plate from the left side of the equipment, and store the dismantled support plate and screws for later use. The specific position of the support plate is shown in the following figure.



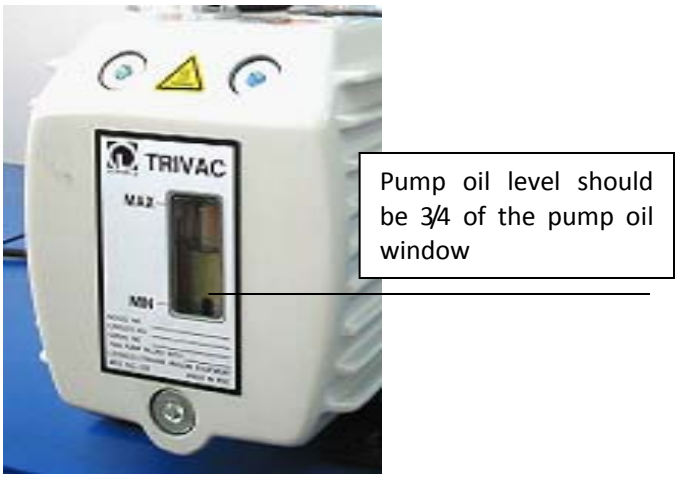
3.2. Check the vacuum pump

1) Dismantle the lower back cover on the back side of the equipment. Please refer to the Back View in the next page for the position of the lower back cover.



Pic 1 Back view

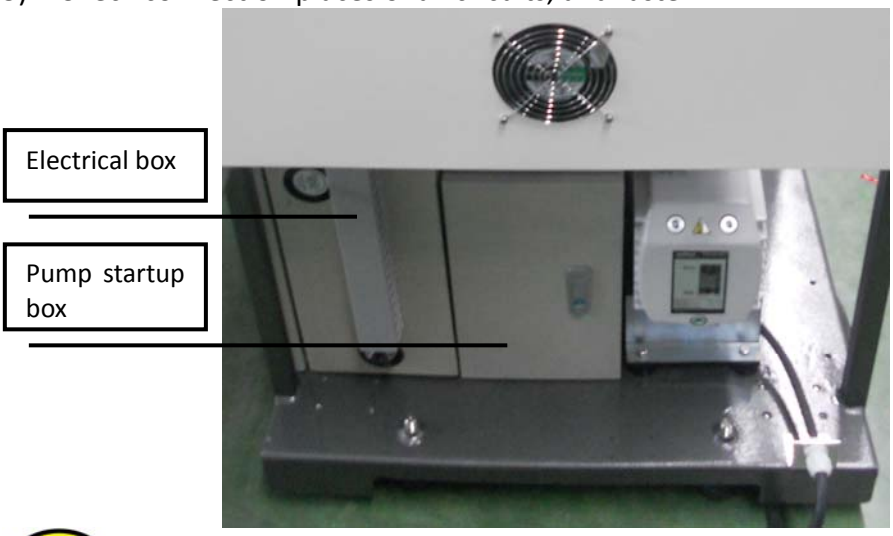
2) Check the vacuum pump oil. The oil level should be not less than 3/4 of the pump oil window (if necessary, supplement the circulating pump oil). Please refer to the View of Vacuum Pump in the next page.
3) Check whether all fixing bolts of the vacuum pump are firm.



Vacuum pump

3.3. Check wires and pipelines

1) Remove the two screws which are located M5* 10mm on the lower bottom of the right cover (see Left View I of Equipment in Appendix IX), and take off the right cover.During dismantling, do not knock against the surface of the cover to avoid scratching the equipment.
2) Check and fasten plasma power wires.
3) Open the electrical appliance box and the pump startup box, checking that all connections are in place.
4) Check hoses in the H_2O_2 supply part, and fasten carefully.
5) Check connection places of all circuits, and fasten.



After the equipment is in position, it is necessary to check wire circuits and pipelines carefully, and fasten one by one.

3.4. Supply power to the equipment and check electric parts

1) Start up the equipment. Connect the equipment to a 380V power supply through a wire, open the air switch panel door on the upper back cover, and turn the circuit breaker to “ON” (flip it upwards). Press the (ON) button on the control panel on the lower front cover.
2) Judge whether the three-phase electric wiring is right. As the vacuum pump of

the equipment is 380V, the vacuum pump can rotate forwards and reversely after being connected with a power supply. Therefore, when the 80V power supply is connected, it is necessary to judge whether the phase sequence of the 380V power supply is right. The judgment method is: after the power supply is connected, open the pump startup box, and judge that wiring is right if the NORMAL indicating lamp of the phase sequence protector (See Appendix VII) lights. It is necessary to adjust the wiring sequence of the power supply if the REVERSAL indicating lamp lights.

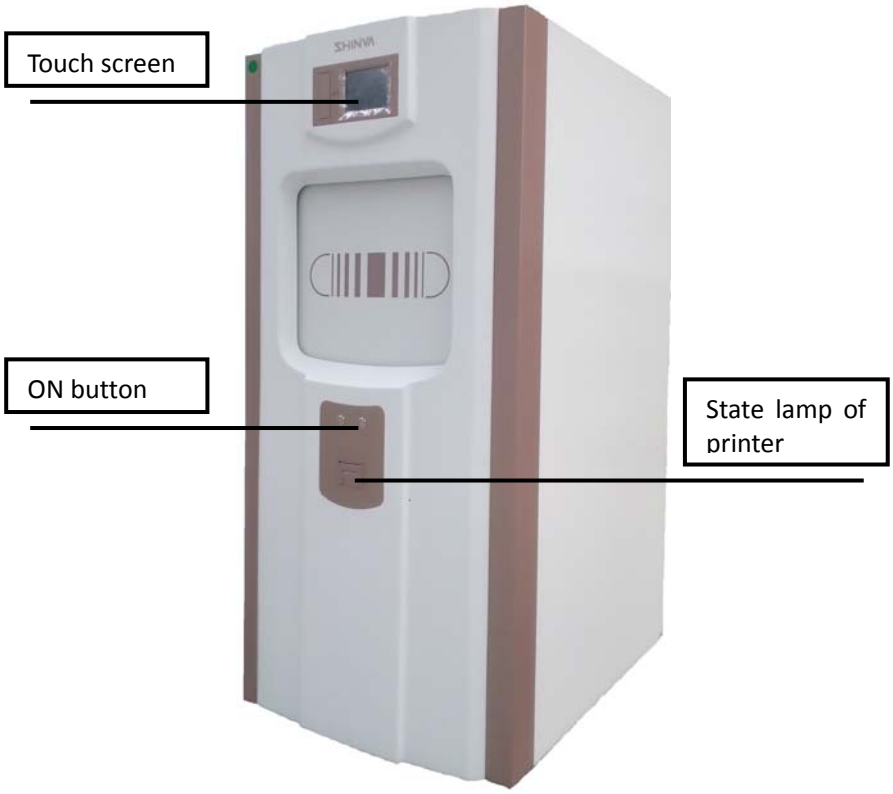
3) Check and adjust the plasma power supply. Check whether the low-voltage switch and high-voltage switch of the plasma power supply are turned on and in the ON position, and when the procedure is run to the lower vacuum limit, after the plasma power supply works, check whether the current value in the current display screen is within 1.3A- 1.5A. If the value is not right, rotate the knob “Power Adjust” to adjust the value.

4) Check whether fans work normally：

- Ventilating fans on the back side and top part of the sterilizer and the lateral walls of the electric box.
- Purifier cooling fans

5) Check all indicators. Please refer to the following figure for the position of indicators.

Type	Indicators	Normal state
1	Touch screen	Show the initial screen
2	ON button	The indicating lamp lights
3	State lamp of printer	The indicating lamp lights



6) Check all control elements. Enter the manual operating interface, and click all valves and pump buttons one by one, to affirm that they work normally.

7) Check door OPERATION. Click the Door OPERATION button to confirm that the sealing door runs without abnormal noise. During door opening/closing, pull down the obstacle switch cover to confirm that the obstacle switch can run normally.
8) Check the printer. Open the cover of the printer to confirm that printing paper is lead outside the printer from the upper part of the whole coil of printing paper. Manually press down the TEST button on the printer to ensure that the printing paper moves smoothly.
9) Check integrated biological incubator (optional configuration). Check whether biological incubator's temperature reaches 56℃, and click each button in turn to check whether each button works normally.

3.5. Check cassette parts

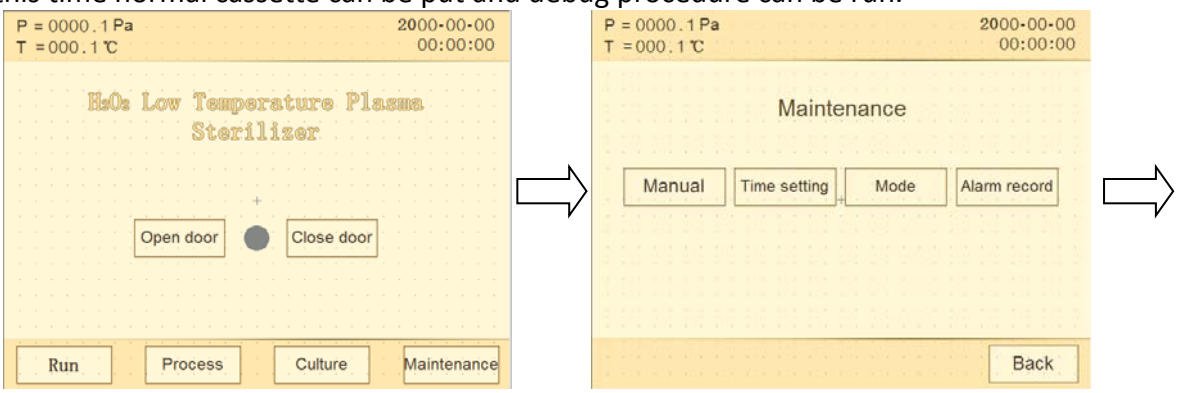
1) Check carefully whether each screw is loose on cassette part, especially for slippery course fixed screw, motor fixed screw, needle, fixed screw for connection plate connecting cassette and motor shaft and fixed screw for cassette and slippery course and slider.
2) Debug cassette's run condition.

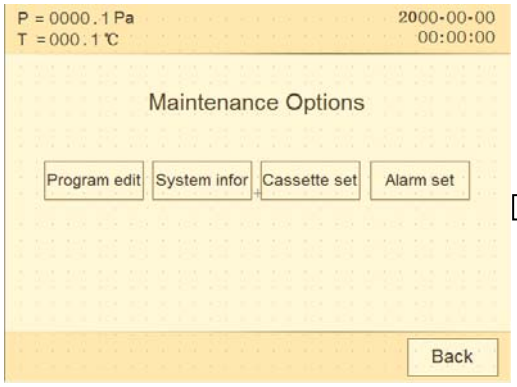
According to prompt on touch screen,， put in and quit Hydrogen peroxide cassette spent out to check whether the cassette part run normally. Specific content are referred to operating manual.

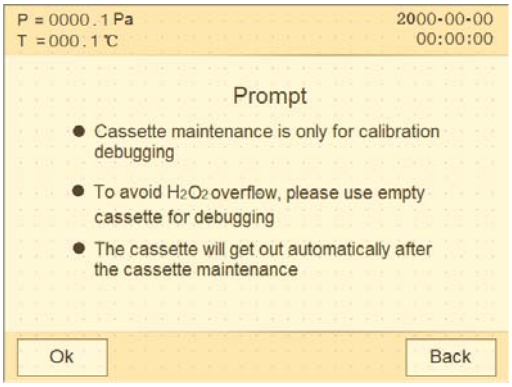
Put in vacant cassette, according to process on the below picture entering cassette maintenance interface, clicking forward and puncture (simulating puncture but not puncturing cassette) in turn, check and confirm whether 12 holes can be punctured normally in turn (needle is within the scope of puncturing hole).

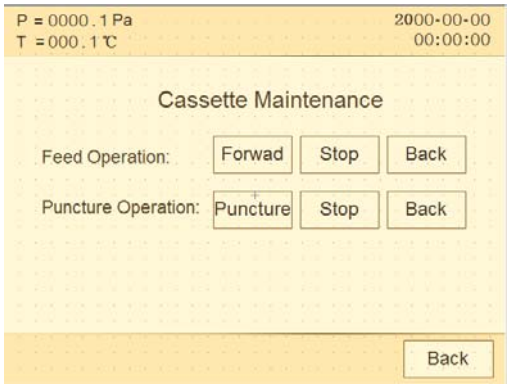
If it doesn't operate correct puncture, needle installment plate height need to be adjusted or cassette fixed subject's levelness need to be adjusted.

After debug is finished, quitting debug interface and cassette quits automatically. At this time normal cassette can be put and debug procedure can be run.









Hydrogen peroxide is a strong oxidizer and can easily penetrate skin, causing temporary damage.

In case of contact with the skin by accident, immediately wash off with plenty of water for about 5 minutes. If inhaled or swallowed, immediately drink plenty of water, and then seek medical treatment.

3.6. Run the procedure

After the above steps are completed, run the holding pressure cycle procedure, standard cycle procedure and enhanced cycle procedure, and observe whether there are any problems, such as warnings or alarms. Examine whether the printing record is normal.

3.7. Check the basket

Check whether the basket has appearance defects caused during transport such as deformation, then open the front door, put the basket in, and check whether the slide action is smooth.

3.8. Installing the outer cover

1) Adjust the circuit breaker on the upper back cover of the equipment to OFF.
2) Fix the lower back cover to the framework by a key.
3) Fix the left cover and the right cover to the framework with M5 screws.

4. Equipment acceptance

According to requirements in Part III: Central Sterile Supply Department Surveillance Standard for Cleaning, Disinfection and Sterilization, in the case of new installation, movement, overhaul, sterilization failure, change of packing materials or changes of articles to be sterilized, the sterilization effect should be reevaluated, including

monitoring by physical monitoring, chemical monitoring and biological monitoring(repetition for 3 times). The sterilizer can be used only after the monitoring has been qualified. The specific methods are as follows:

Article loading

Fill loads not more than 80% of the volume in the sterilizing chamber. The loads should be packaged with paper-plastic packaging bags or non-woven bags (the specific loading requirements can be found in the Operation Manual). An intra-bag chemical index card should be put in the hardest to sterilize position inside each bag. After packaging three self-contained biological indicators with paper plastic packaging bags or non-woven bags, put them in the front, middle and back three positions of the basket inside the sterilizing chamber.

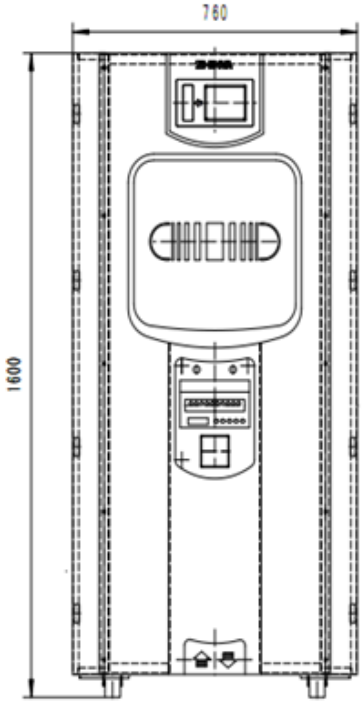
Operation procedures

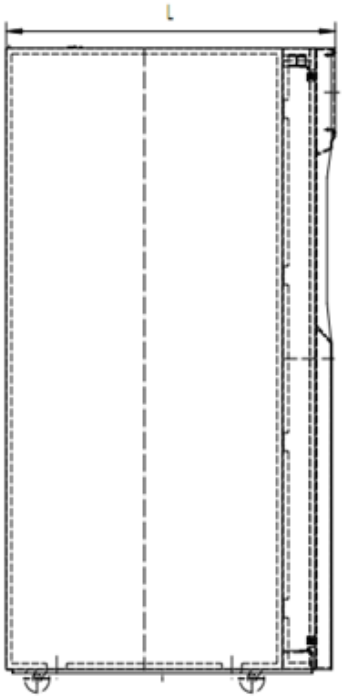
Run the standard cycle procedure after loading articles.

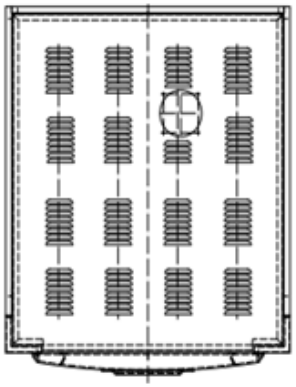
Product release

After the printing record is viewed to prove the qualification of physical monitoring; the color change of the intra-bag chemical index card proves the qualification of chemical monitoring; the qualification of biological cultivation proves the qualification of biologic monitoring, and the product is run for three times continuously, the equipment can be said to be normally.

Appendix 1. Installation diagram of PS-100/200 (single-door)



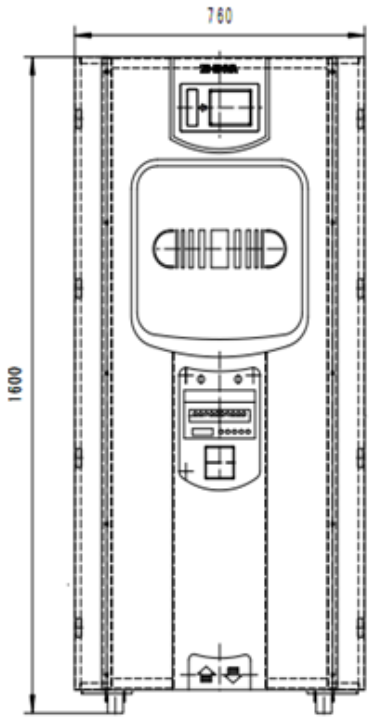


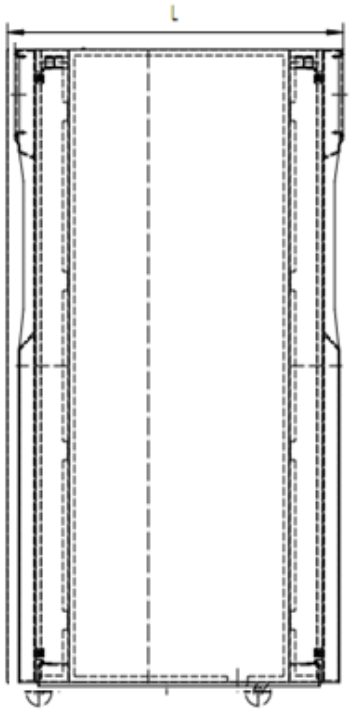


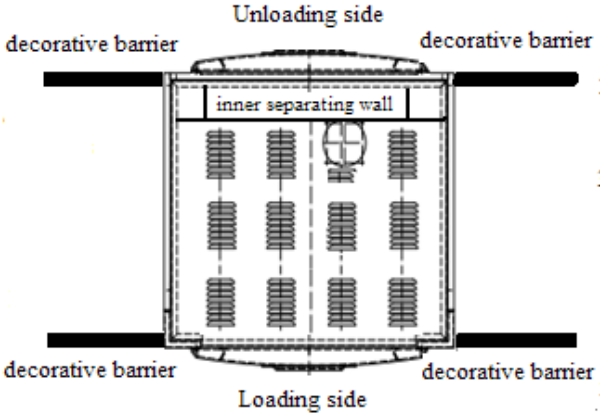
Technical requirements

1. PS-100:chamber size:750x450x400mm(LxWxH)
   Overall size:990x760x1600mm(LxWxH) weight:400kg
   PS-200:chamber size:1250x450x400mm(LxWxH)
   Overall size:1470x760x1600mm(LxWxH) weight:500kg
2. Indoor installation. Temp:10~35℃ Relative humidity: 30%~75%. Atmosphenc pressure: 70~106KPa
   Floor flatness:±5mm/m
3. Distance between wall and equipment both and back sides:100mm.
   PS-100:distance between operation side and obstacle should> 1m.
   PS-200:distance between operation side and obstacle should> 1.5m.
4. Maintenance space: at least 1m around the equipment.
5. Power voltage:380V±38V. PS-100: 3KW, 12A
   PS-200: 3.5KW, 15A

Appendix 2. Installation diagram of PS-100/200(double-door)



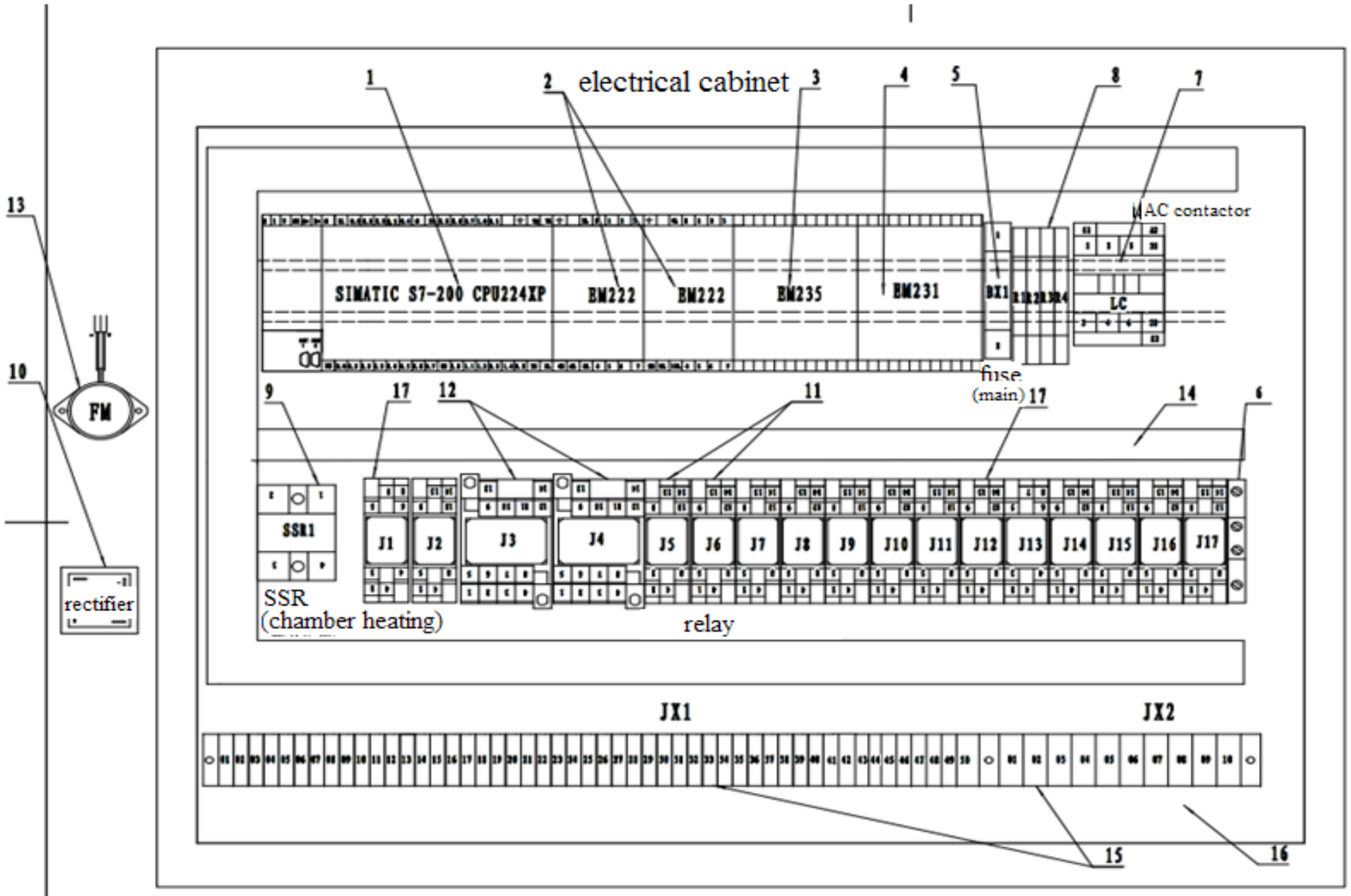




Technical requirements

1. PS-100:chamber size:750x450x400mm(LxWxH)
   Overall size:990x760x1600mm(LxWxH) weight:400kg
   PS-200:chamber size:1250x450x400mm(LxWxH)
   Overall size:1470x760x1600mm(LxWxH);weight:500kg
2. Indoor installation. Temp:10~35℃ Relative humidity: 30%~75%. Atmospheric pressure:70~106KPa
   Floor flatness:±5mm/m
3. Distance between wall and equipment both and back sides:100mm.
   PS-100:distance between operation side and obstacle should＞1m.
   PS-200:distance between operation side and obstacle should＞1.5m.
4. Maintenance space: at least 1m around the equipment.
5. Power voltage:380V±38V. PS-100: 3KW, 12A
   PS-200: 3.5KW, 15A

Appendix 3. Layout diagram of electric box main board



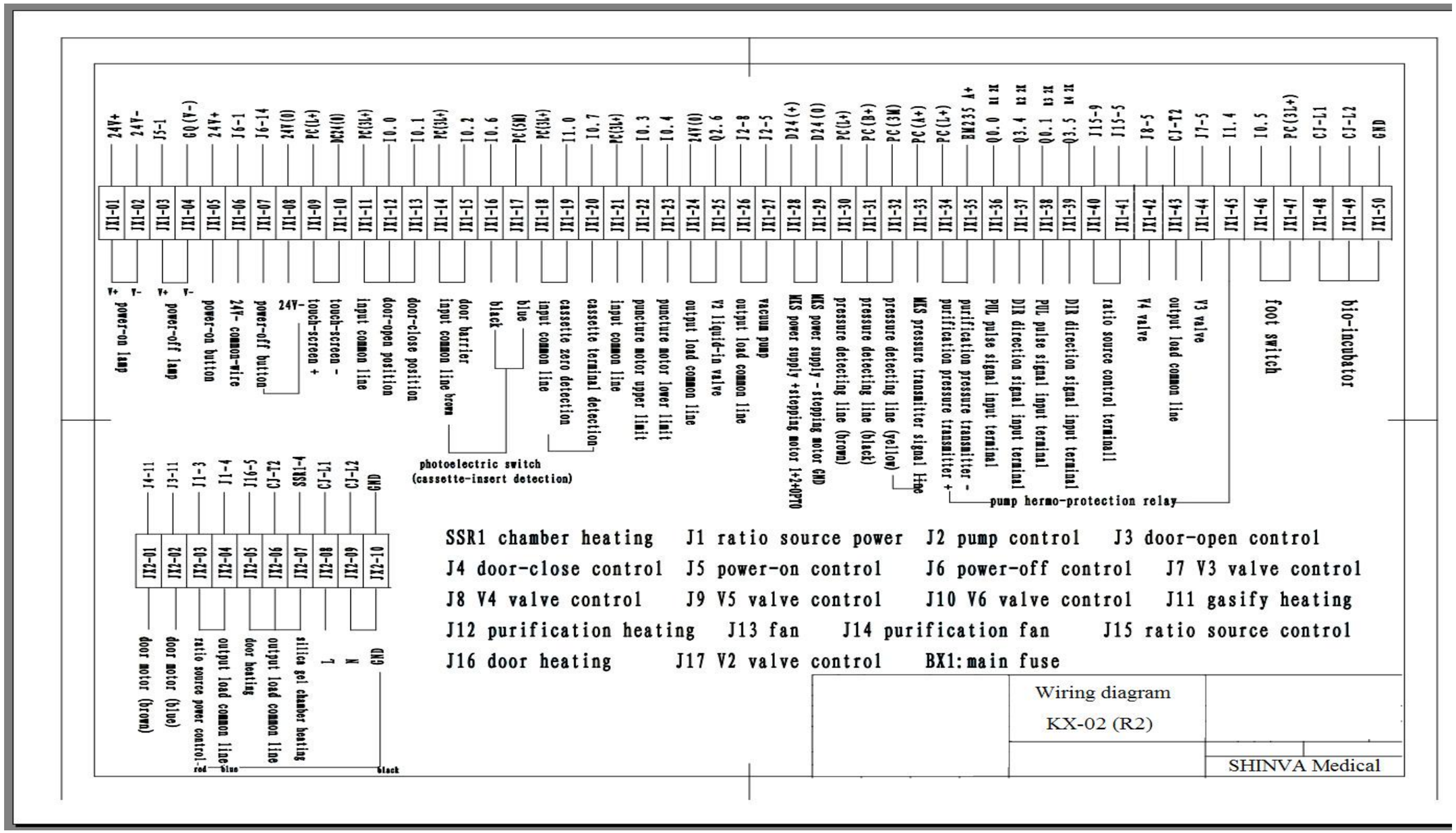
No	Component name
1	CPU224XP
2	EM222 extended module
3	EM235 extended module
4	EM 231 extended module
5	main fuse(20A)
6	terminal
7	AC contactor
8	plastic terminal
9	SSR
10	rectifier bridge
11	5A relay
12	10A relay
13	buzzer
14	wiring duct
15	plastic terminal
16	circuit board
17	12A relay

Note: J1,J2 are LY2NJ 12A relay; J3,J4 are 10A relay; the rest are LN2NJ 5A relay

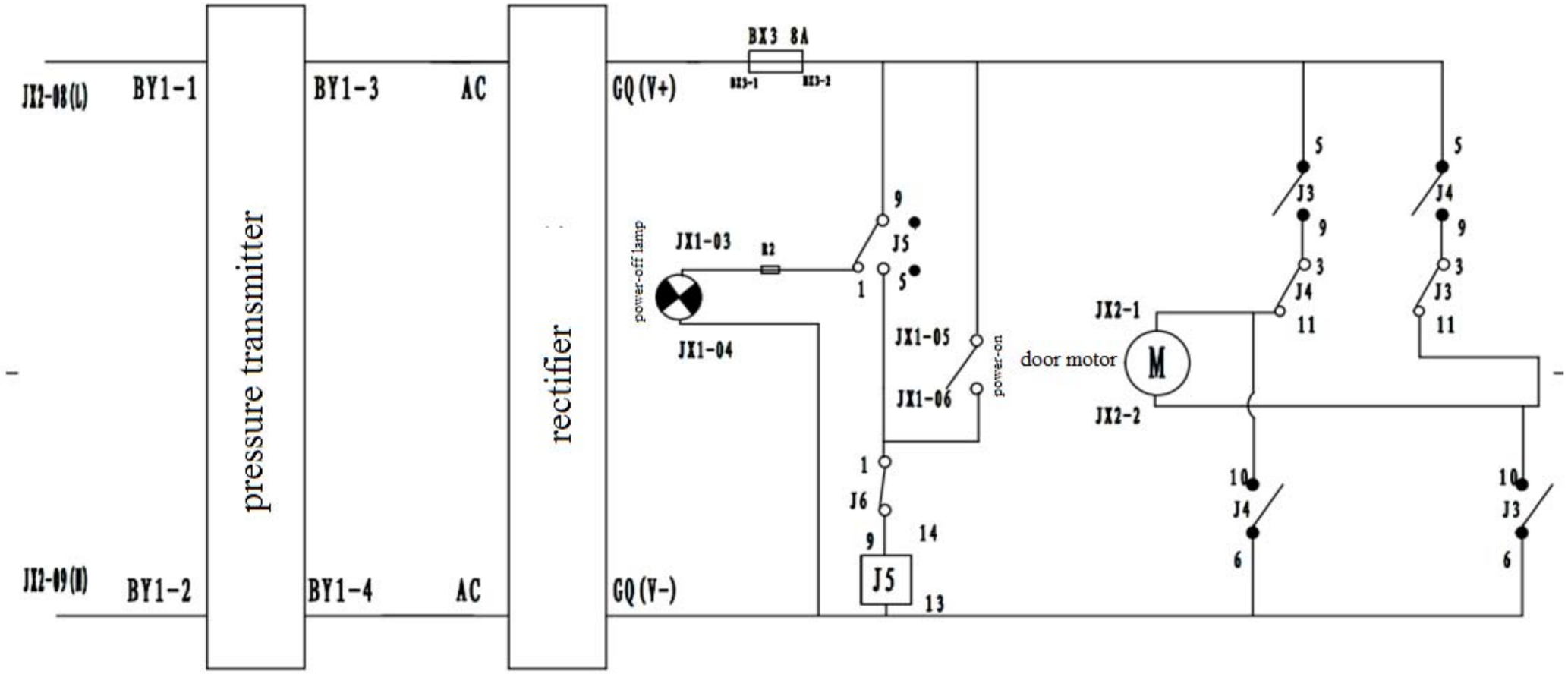
Installation procedure of electrical board (on the cabinet door):DC24V power,gateway storage,BH24 door motor protection and temp transmitter.

Make sure the installation upward to reserve enough space for wiring terminals.

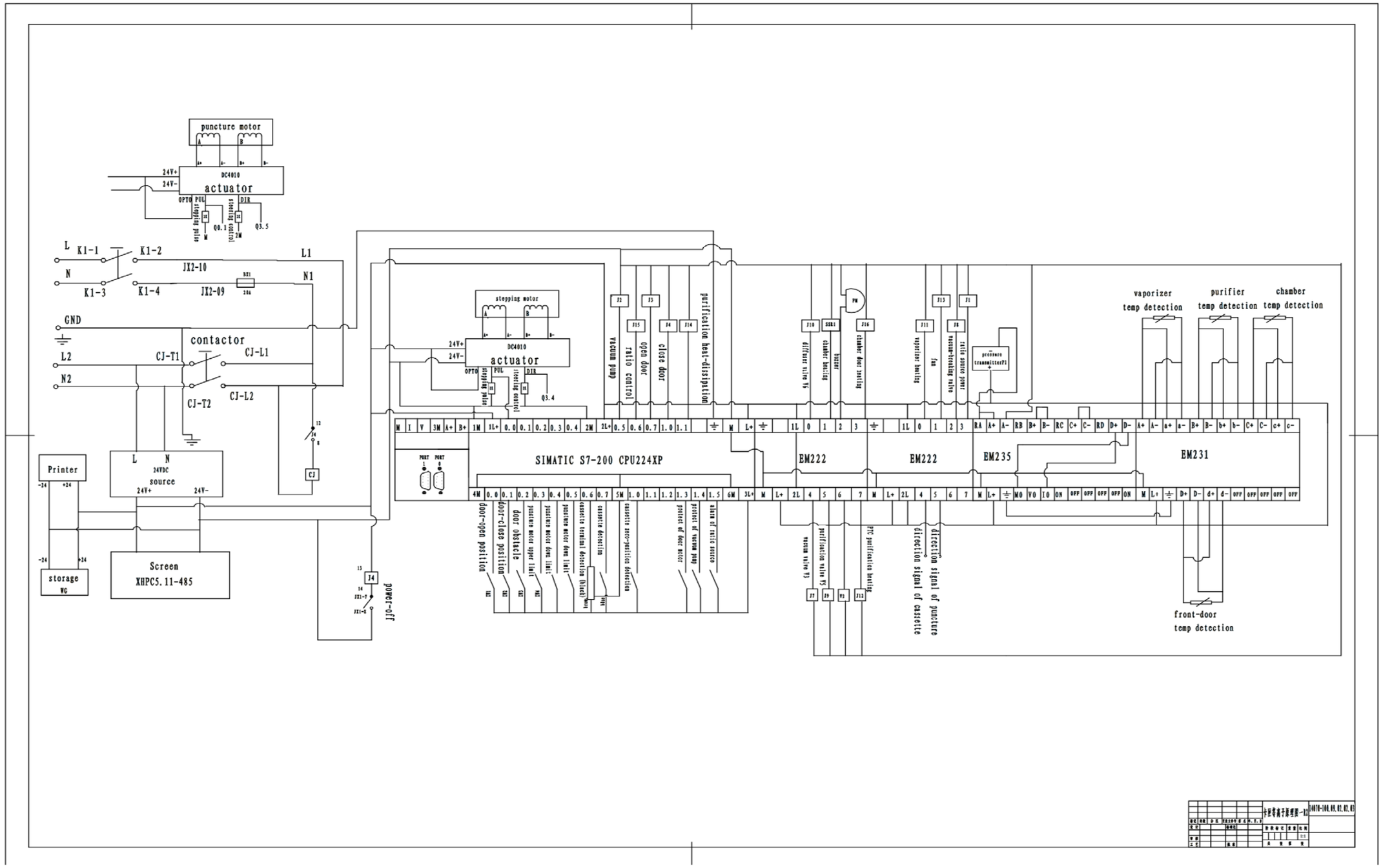
Appendix 4. Wiring diagram of electric box



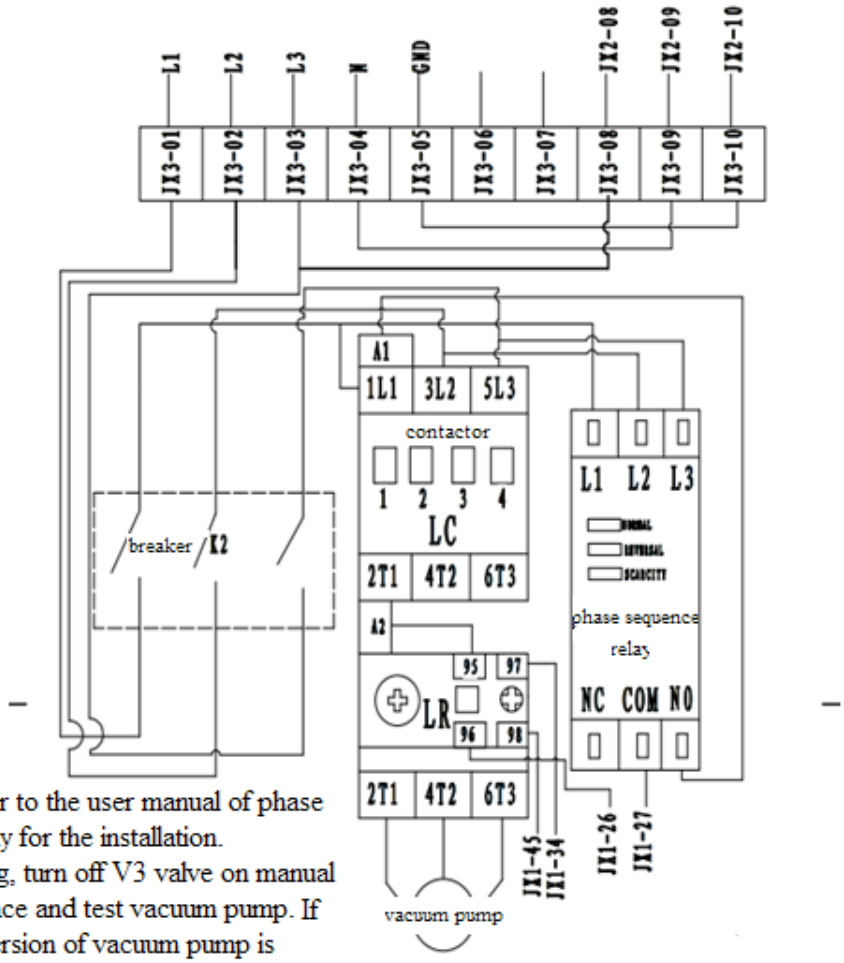
Appendix 5. Transformation, rectification and power-on schematic



Appendix 6. Electrical circuit sketch



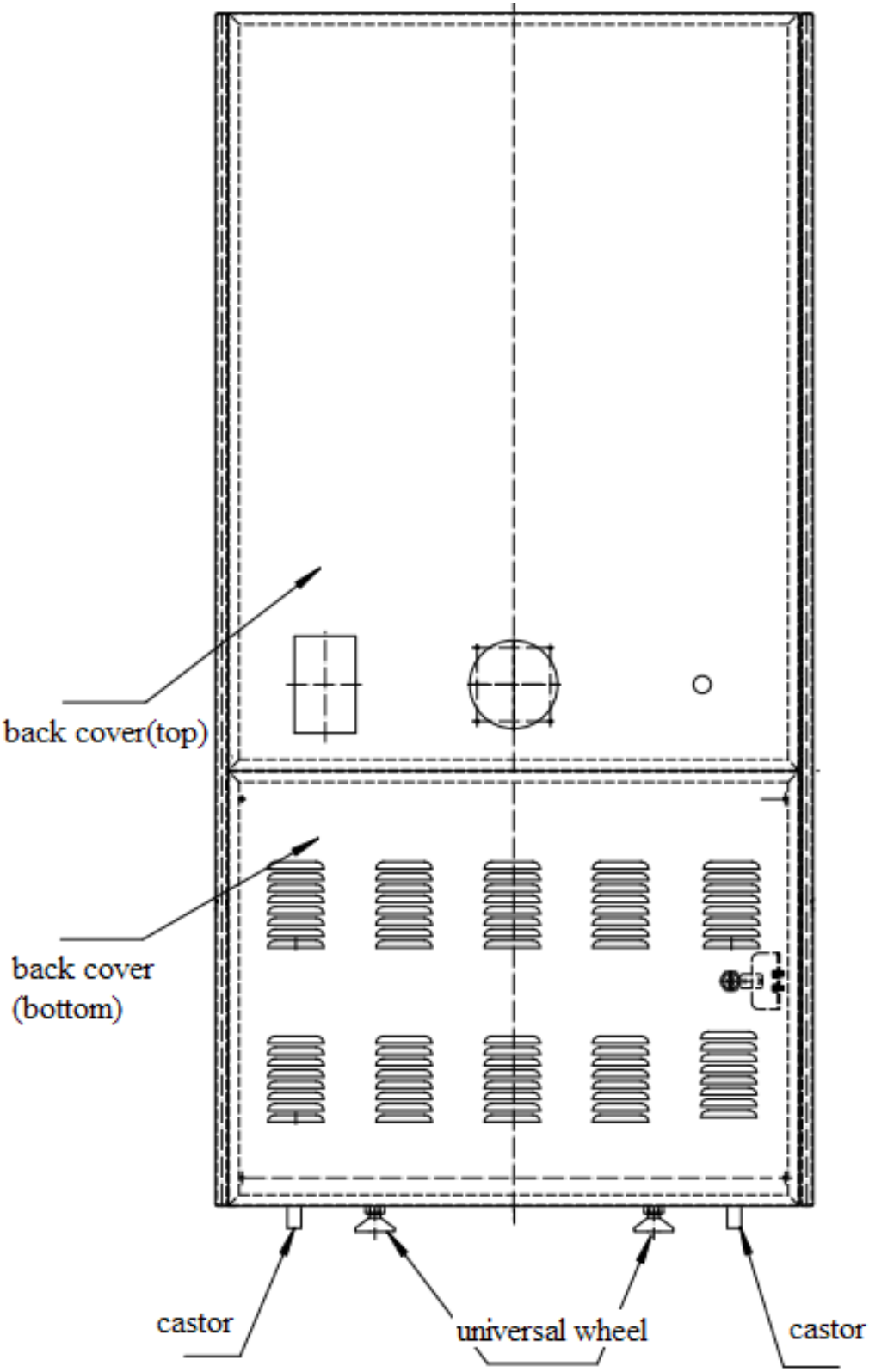
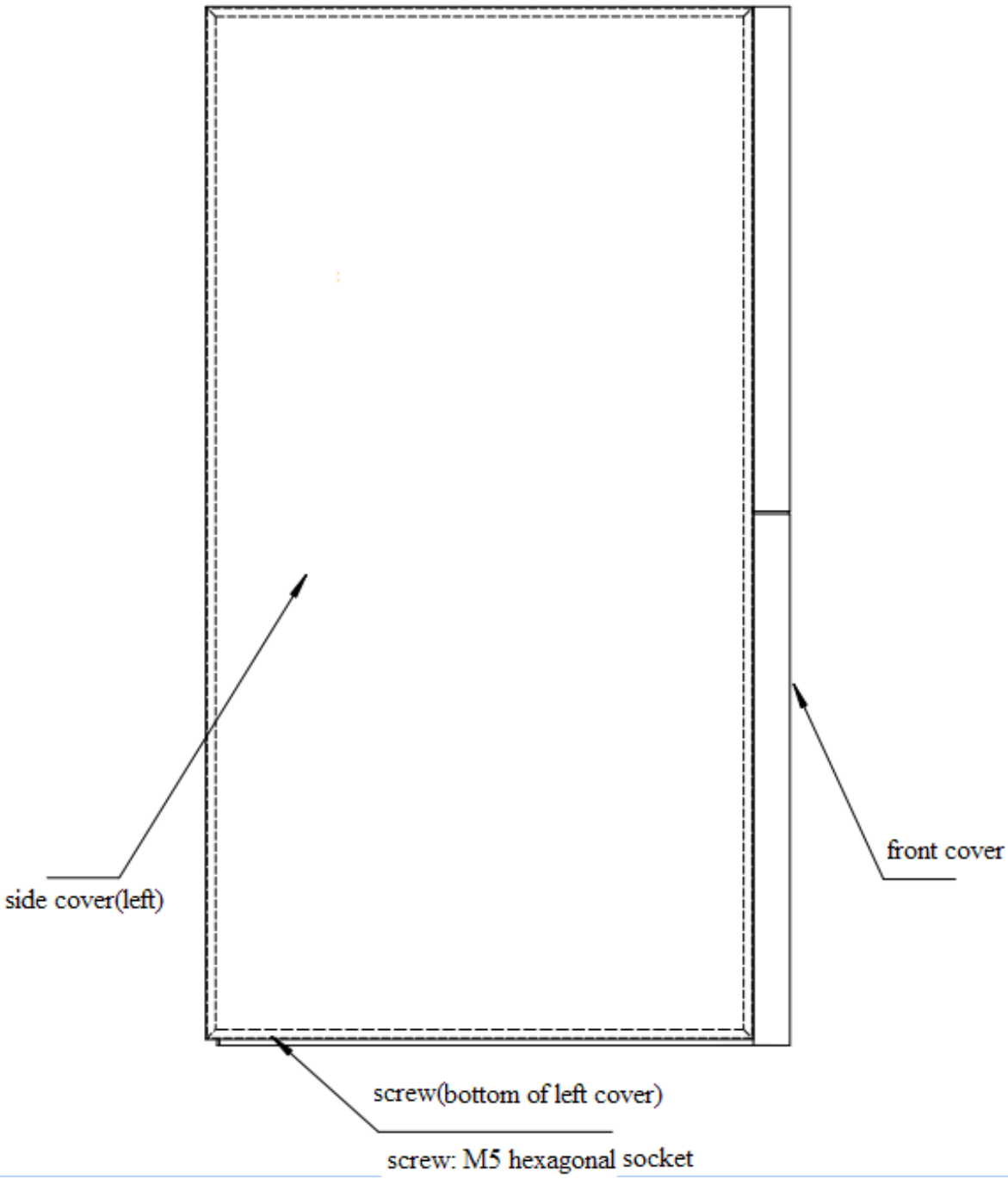
Appendix 7. Wiring diagram of pump startup box



Notes:

1. Please refer to the user manual of phase sequence relay for the installation.
2. After wiring, turn off V3 valve on manual control interface and test vacuum pump. If the phase inversion of vacuum pump is correct and the green light of phase sequence relay is on, that indicates wiring is correct.
3. If the green light of phase sequence relay is on but the phase inversion of cacuum pump is wrong, please change any 2 phases of L1,L2 and L3 of vacuum pump wiring.
4. If other matters cause the wrong phase sequency of the main power, the protection will be activated during the vacuum pump operation, and the yellow light will be on. Now the vacuum pump stops working and the phase sequency can be changed.

Appendix 8. Bottom view of equipment



Appendix 9. Left side of equipment